



Postgraduate Institute of Medical Education and Research, Chandigarh

Department of Hematology

Flow cytometric assay for minimal residual disease (MRD)

Name V K Gurg Age/sex 65 Yr/M CR No. 201405136462
Ward/OPD Clinical Hematology & Medical Oncology-Hematology Clinic

Requisition Date	Lab ID	Sample	Consultant	BMA No.
07/12/2022	MRD-828/22	Bone Marrow Aspirate	-	A-2960/22

TLC (x10 ⁹ /L)	Neutrophils (%)	Lymphocytes (%)	Monocytes (%)	Eosinophils (%)	Other (%)
12.5	-	-	-	-	-

Bone Marrow Diagnosis

-

Gated events

CD138 vs CD38 (JK36 nanobody) vs CD45 vs side scatter - Nil

RESULT

Percentage of MRD (%) Nil
Number of events acquired 1.6 million
Cellularity of the sample Adequate
Viability of the sample (%) 92.32%

Comments

No plasma cells could be gated using CD38 (JK36 nanobody) and CD138.

Advice

-

Senior Resident Dr. Anjali Faculty Dr. Nabhajit Mallik
Typed by Mr. Gurpreet Singh Validated on 14-12-22 12:24 PM
Technologist Mrs. Parveen Bose Validated by Nabhajit Mallik(Assistant Professor)



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Postgraduate Institute of Medical Education & Research, Chandigarh
REPORT OF BONE MARROW ASPIRATION / TREPHINE BIOPSY

Name: V K Gurg **Age / Sex:** 65 Yr/M **CR No:** 201405136462 **B.M.No:** A-2960/22
Cl.I/c: **Ward:** Clinical Hematology & Medical Oncology- **Dated:** 07/12/2022
Hematology Clinic

Clinical K/C/O- Multiple myeloma on therapy- post 6 cycles of Dara-VCD. Now planned for hematopoietic
Diagnosis:- stem cell transplant. Bone marrow for MRD assessment.

HEMOGRAM DETAILS

Hemogram No: 120

HB:- 12.3 **gm/dl** **Retic:** 1.07 **%** **PLT:** 148 **X10⁹/L** **TLC:** 12.5 **X10⁹/L**
DLC **P-** 77 **L-** 07 **M-** 14 **E-** 02 **Ba-** - **BI-** - **Pm--** **My-** - **Mm--** **nRBC-** -

PBF: Normocytic normochromic red blood cells with a few macrocytes seen. Platelet are adequate.

BONE MARROW FINDINGS

Particles: Particulate **NE:E Ratio** - **M:E Ratio** 3.2:1

Cellularity: Normocellular to hypocellular.

Blasts: - **Erythropoiesis:**
Promyelocytes: 01 Normoblastic
Myelocytes: 15 to mild megaloblastic.
Metamyelocytes: 06 **Thrombopoiesis:**
Polymorphs: 38 Adequate
Lymphocytes: 16 -
Monocytes; - **Others:**
Eosinophils: 01 Erythroid precursors- 20%, Eo baso precursors-01%.
Basophils: -
Plasma Cells: 02

CYTOCHEMISTRY

LAP - **MPO** - **PAS** - **PERLS** 2+

TREPHINE BIOPSY REPORT

Trephine Biopsy No :Tx- 2471/22

Bilateral trephine biopsies measuring 2 cm and 2.5 cm shows variably cellular marrow spaces ranging from markedly hypocellular (cellularity= less than 5%) to normocellular (cellularity= 30-40%) marrow spaces with an overall cellularity of 20-30%. The cellular marrow spaces seen show proportionate representation of all the three hematopoietic lineage elements.

Reticulin - -

Interpretation K/C/O- Multiple myeloma on therapy- shows variably cellular bone marrow and 02% plasma cells (details as above).

Advice (i) Kindly collect and correlate with flow cytometry MRD report (available online). (ii)
Correlate with serum/urine protein electrophoresis and other parameters.

JR Dr. Suruthy **SR** Dr. Sunny Bhardwaj **Faculty** Dr. Shano Naseem
Validated On 10-12-22 04:01 PM **Validated By** Shano Naseem(Professor)



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